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(54) **APPARATUSES AND METHODS FOR
FACILITATING PLATFORM-AGNOSTIC
NETWORK FUNCTIONS**

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(57)

ABSTRACT

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Related U.S. Application Data

(63) Continuation of application No. 17/480,245, filed on
Sep. 21, 2021, now Pat. No. 12,333,320.

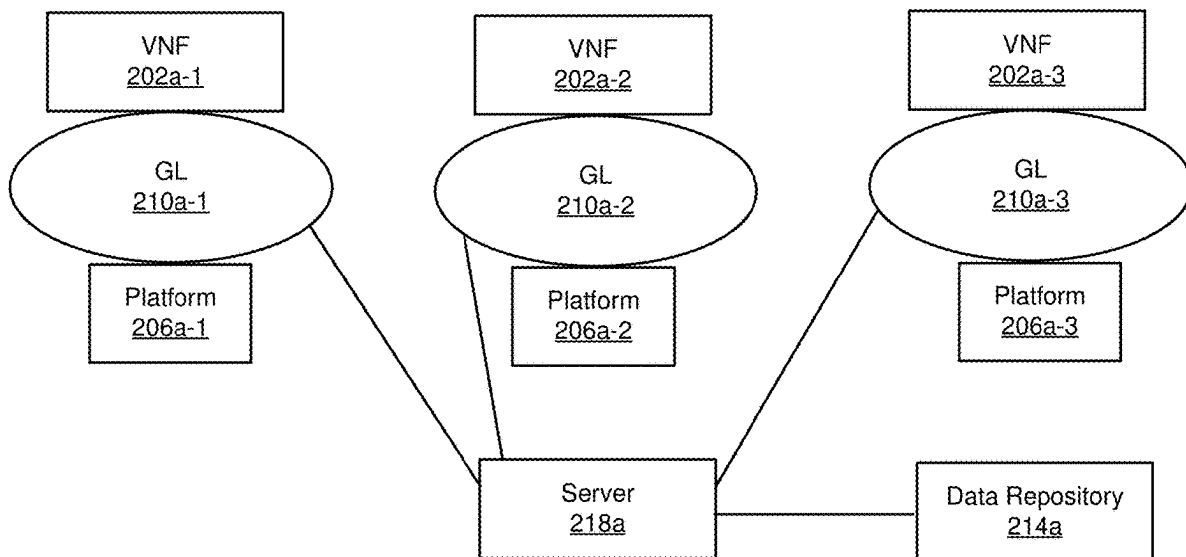
Publication Classification

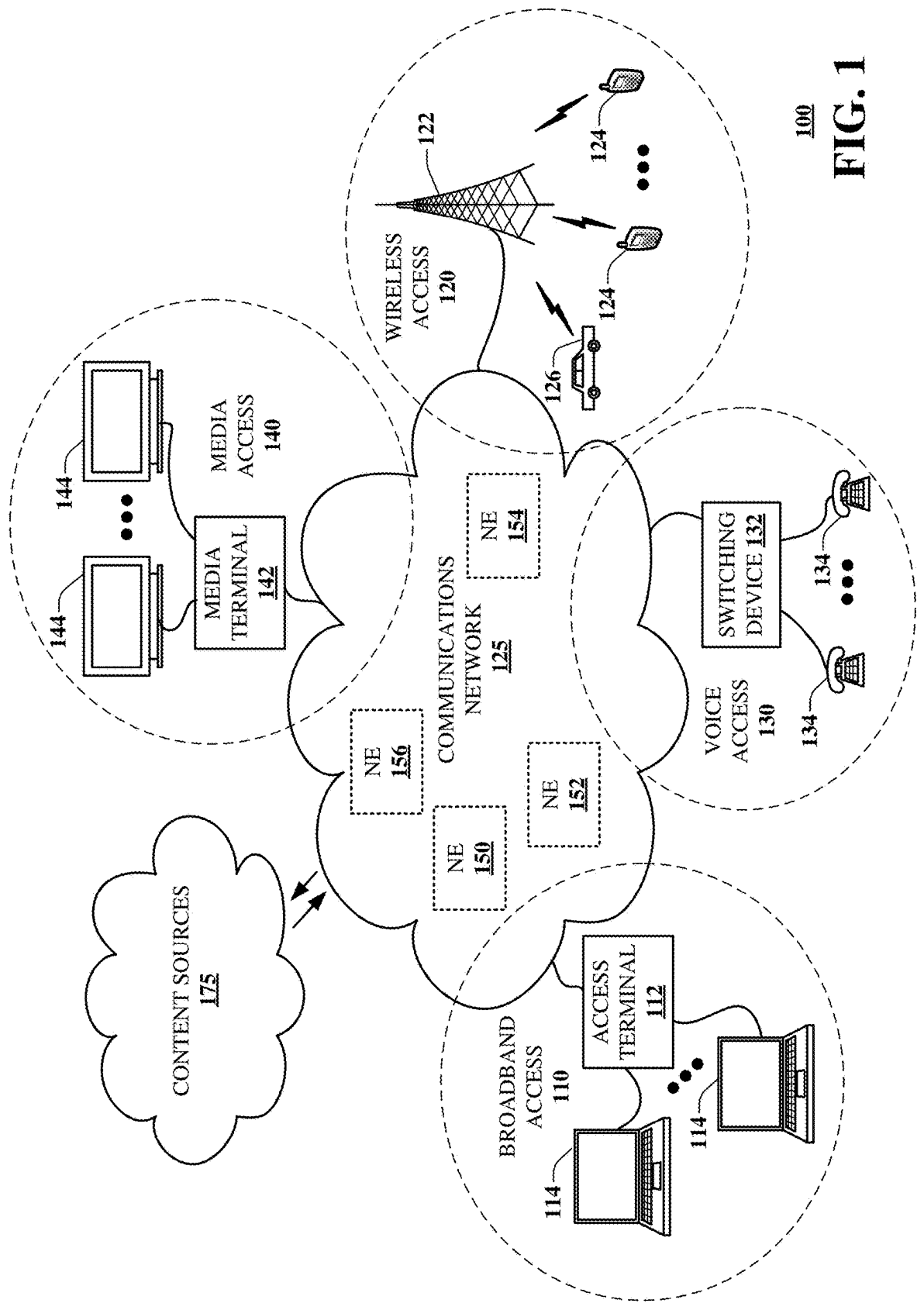
(51) **Int. Cl.**

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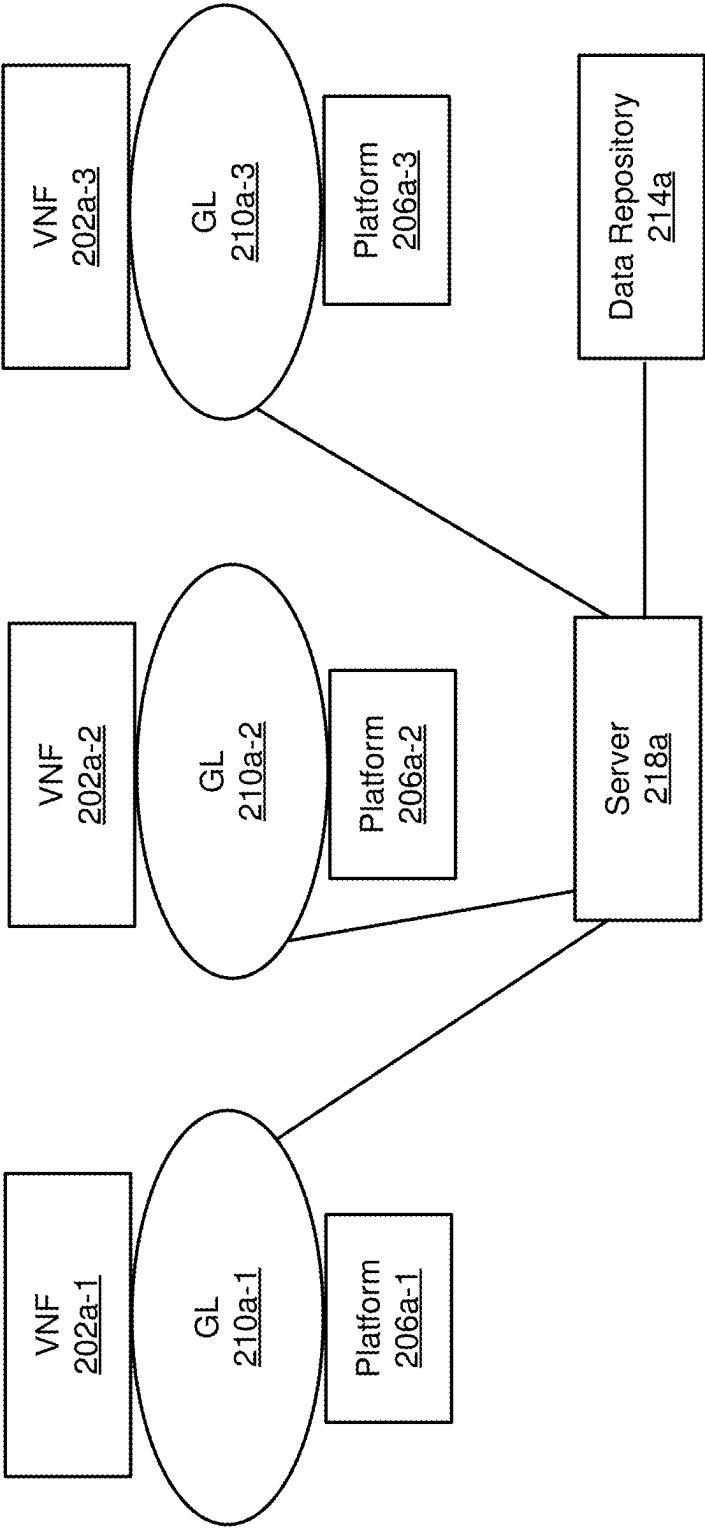
G06F 9/50 (2006.01)

Aspects of the subject disclosure may include, for example,
coupling a virtual network function (VNF) to a cloud
computing platform, resulting in an instantiation of the VNF
with respect to the cloud computing platform, obtaining,
based on the coupling, data to operatively interface the VNF
and the cloud computing platform, and translating a com-
munication, a message, or a combination thereof, between
the VNF and the cloud computing platform using the data.
Other embodiments are disclosed.

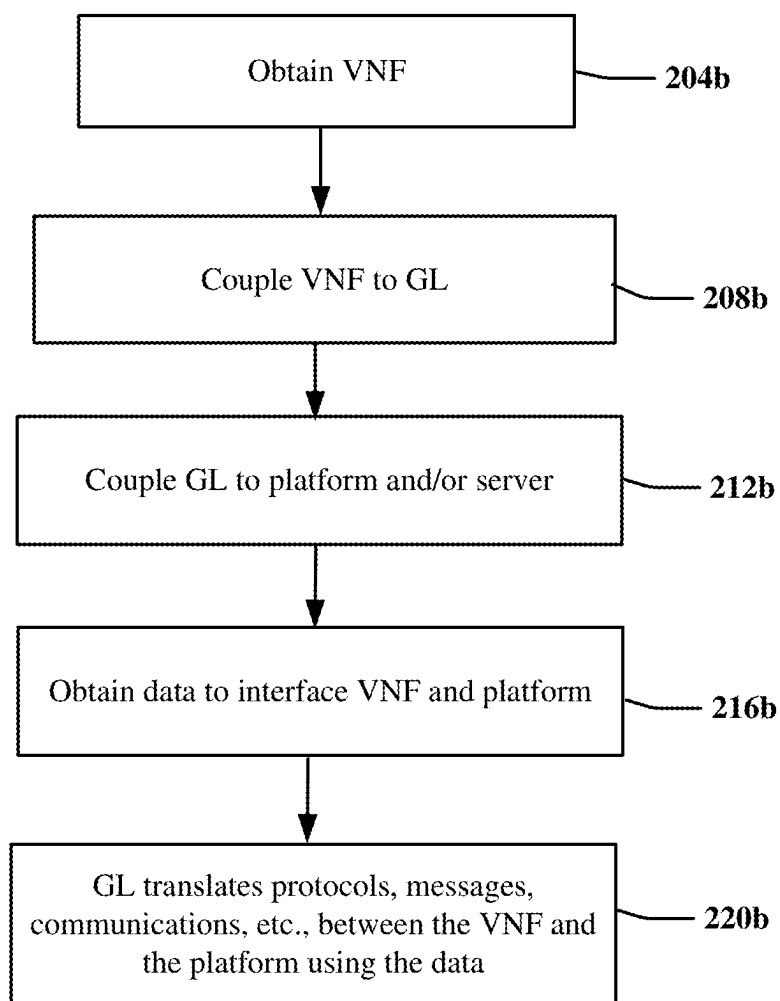




100
FIG. 1

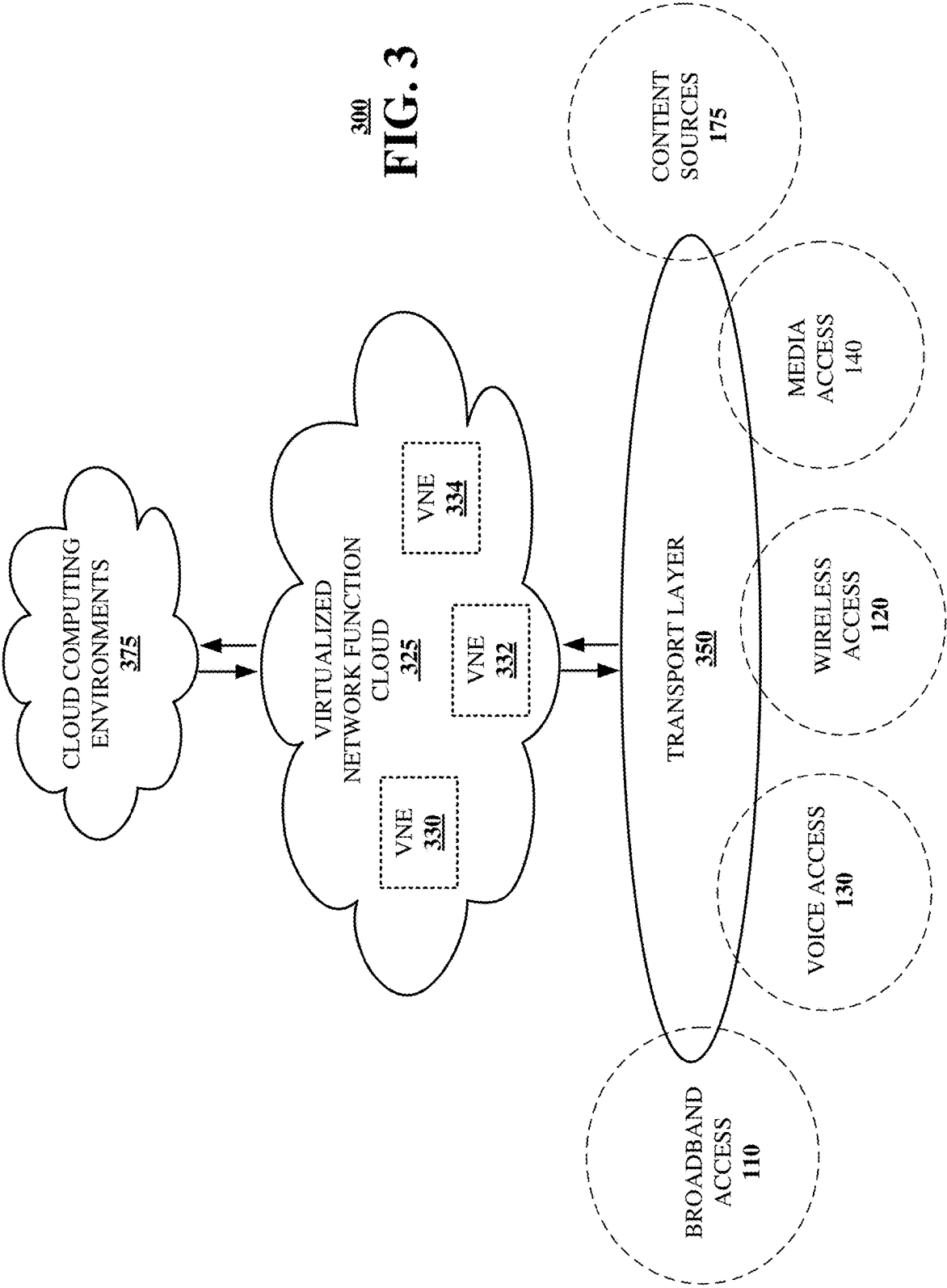


200a
FIG. 2A



200b

FIG. 2B



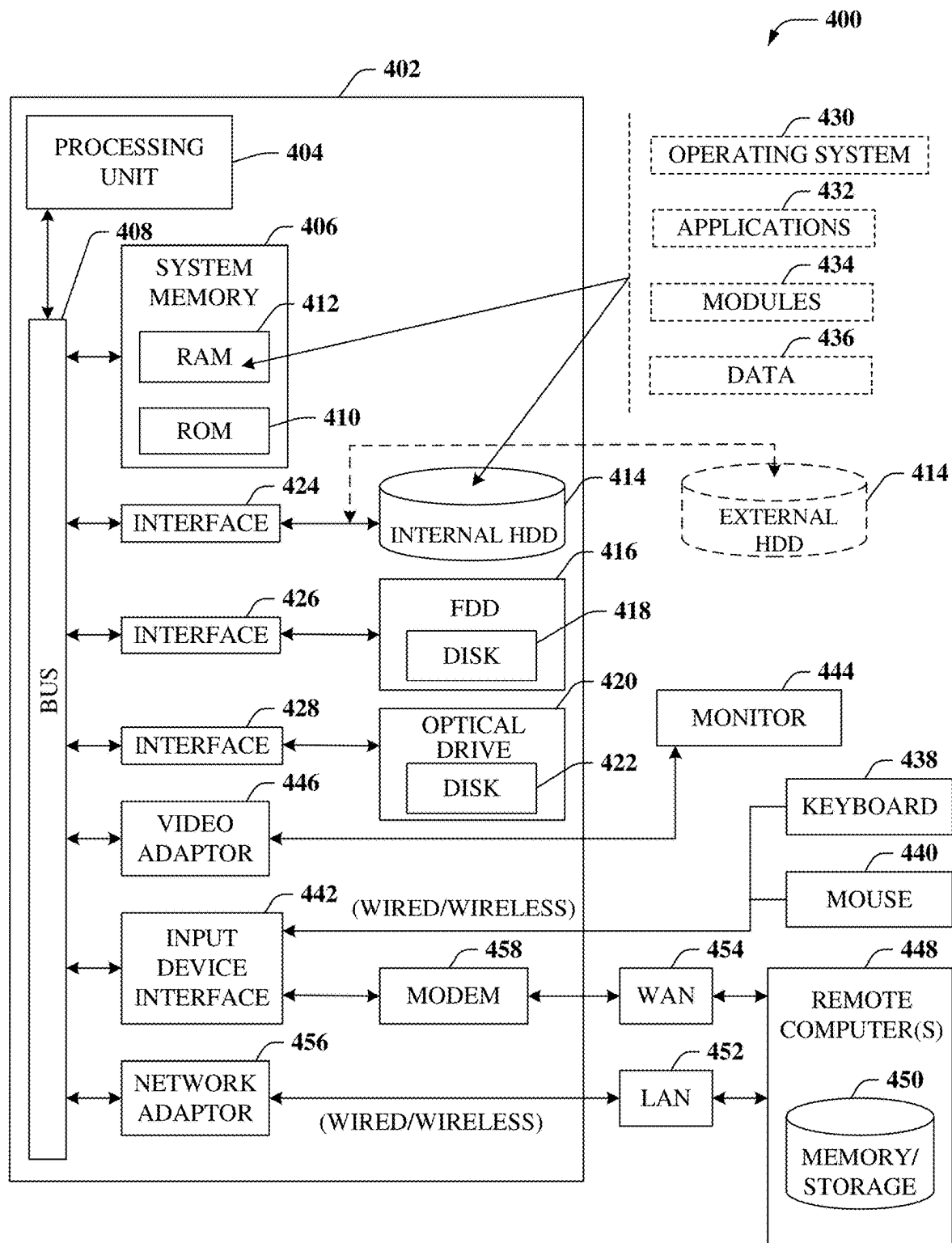


FIG. 4

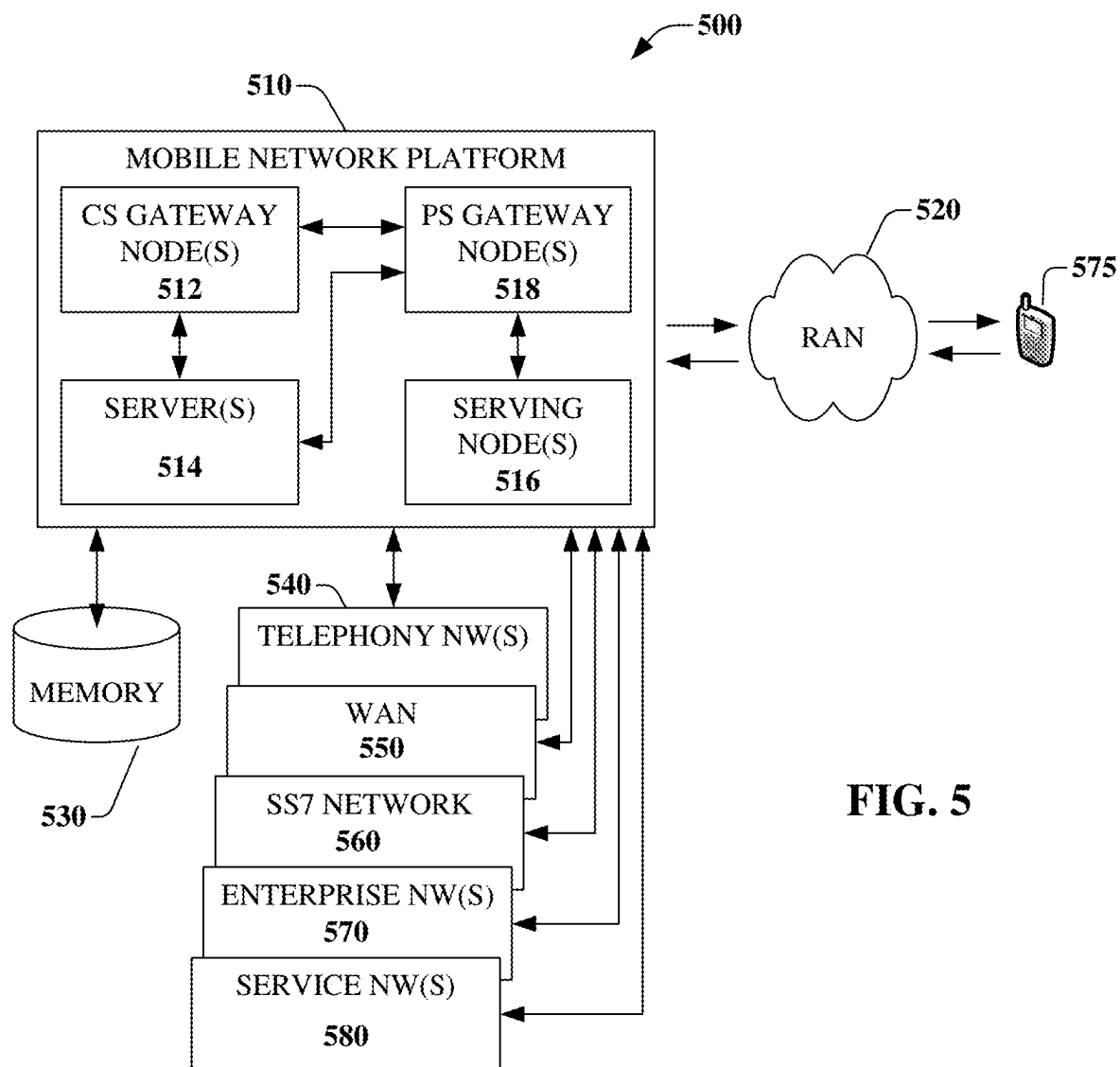
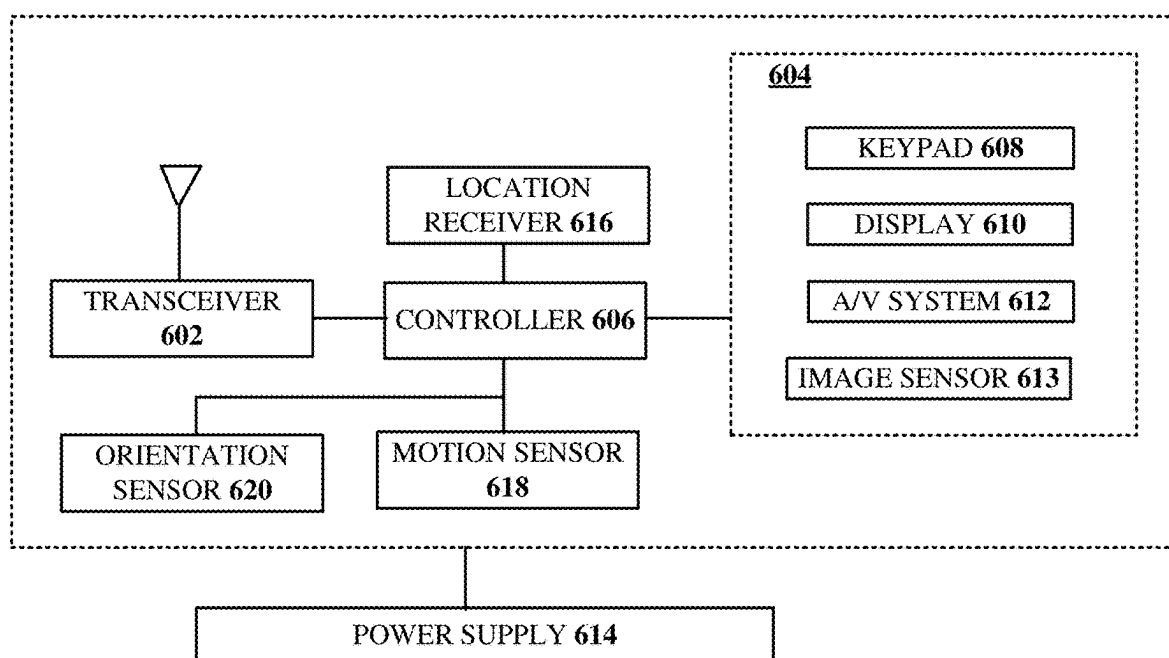


FIG. 5



600
FIG. 6

APPARATUSES AND METHODS FOR FACILITATING PLATFORM-AGNOSTIC NETWORK FUNCTIONS

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] The instant application is a continuation of U.S. patent application Ser. No. 17/480,245, filed on Sep. 21, 2021. All sections of the aforementioned application(s) are incorporated herein by reference in their entirety.

FIELD OF THE DISCLOSURE

[0002] The subject disclosure relates to apparatuses and methods for facilitating platform-agnostic network functions.

BACKGROUND

[0003] Virtual network functions (VNFs), such as virtual machines, containers, virtual functions, and the like, are used in connection with various applications and environments. For example, VNFs may be used in relation to a provisioning of communication services in respect of one or more networks.

[0004] In theory, VNFs may be designed to be coupled to any hardware platform. However, in practice and in the absence of an enforced standard, each platform frequently utilizes its own unique suite/set of configuration parameters, protocols, naming conventions, etc., that makes migrating a given VNF from a first platform to a second platform difficult. For example, migrating the given VNF from the first platform to the second platform may be accompanied by labor-intensive and costly redevelopment, testing, and validation activities. Still further, the cost/expense associated with the migration often results in a VNF developer or provider/developer sticking/staying with a given platform (e.g., the first platform in the foregoing example), imposing risk to the VNF provider/developer in the form of potential platform down-time and service unavailability.

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] Reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

[0006] FIG. 1 is a block diagram illustrating an exemplary, non-limiting embodiment of a communications network in accordance with various aspects described herein.

[0007] FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a system functioning within the communication network of FIG. 1 in accordance with various aspects described herein.

[0008] FIG. 2B depicts an illustrative embodiment of a method in accordance with various aspects described herein.

[0009] FIG. 3 is a block diagram illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein.

[0010] FIG. 4 is a block diagram of an example, non-limiting embodiment of a computing environment in accordance with various aspects described herein.

[0011] FIG. 5 is a block diagram of an example, non-limiting embodiment of a mobile network platform in accordance with various aspects described herein.

[0012] FIG. 6 is a block diagram of an example, non-limiting embodiment of a communication device in accordance with various aspects described herein.

DETAILED DESCRIPTION

[0013] The subject disclosure describes, among other things, illustrative embodiments for instantiating network functions (NFs) across one or more platforms. Other embodiments are described in the subject disclosure.

[0014] One or more aspects of the subject disclosure include, in whole or in part, obtaining a VNF, coupling the VNF to an interface layer, selecting a computing platform from a plurality of computing platforms, resulting in a selected computing platform, coupling the interface layer to the selected computing platform, resulting in an instantiation of the VNF with respect to the selected computing platform, obtaining data to operatively interface the VNF and the selected computing platform, and translating, by the interface layer, a communication, a message, or a combination thereof, between the VNF and the selected computing platform using the data.

[0015] One or more aspects of the subject disclosure include, in whole or in part, coupling a VNF to a cloud computing platform, resulting in an instantiation of the VNF with respect to the cloud computing platform, obtaining, based on the coupling, data to operatively interface the VNF and the cloud computing platform, and translating a communication, a message, or a combination thereof, between the VNF and the cloud computing platform using the data.

[0016] One or more aspects of the subject disclosure include, in whole or in part, coupling, by a processing system including a processor, a network function (NF) to a first computing platform that is operative in accordance with a first plurality of parameters, resulting in a first instantiation of the NF, coupling, by the processing system, the NF to a second computing platform that is operative in accordance with a second plurality of parameters, resulting in a second instantiation of the NF, wherein the second plurality of parameters is at least partially different from the first plurality of parameters, obtaining, by the processing system, first data to facilitate the first instantiation of the NF, and obtaining, by the processing system, second data to facilitate the second instantiation of the NF.

[0017] Referring now to FIG. 1, a block diagram is shown illustrating an example, non-limiting embodiment of a system **100** in accordance with various aspects described herein. For example, system **100** can facilitate in whole or in part obtaining a VNF, coupling the VNF to an interface layer, selecting a computing platform from a plurality of computing platforms, resulting in a selected computing platform, coupling the interface layer to the selected computing platform, resulting in an instantiation of the VNF with respect to the selected computing platform, obtaining data to operatively interface the VNF and the selected computing platform, and translating, by the interface layer, a communication, a message, or a combination thereof, between the VNF and the selected computing platform using the data. System **100** can facilitate in whole or in part coupling a VNF to a cloud computing platform, resulting in an instantiation of the VNF with respect to the cloud computing platform, obtaining, based on the coupling, data to operatively interface the VNF and the cloud computing platform, and translating a communication, a message, or a combination thereof, between the VNF and the cloud com-

puting platform using the data. System 100 can facilitate in whole or in part coupling a NF to a first computing platform that is operative in accordance with a first plurality of parameters, resulting in a first instantiation of the NF, coupling the NF to a second computing platform that is operative in accordance with a second plurality of parameters, resulting in a second instantiation of the NF, wherein the second plurality of parameters is at least partially different from the first plurality of parameters, obtaining first data to facilitate the first instantiation of the NF, and obtaining second data to facilitate the second instantiation of the NF.

[0018] In particular, in FIG. 1 a communications network 125 is presented for providing broadband access 110 to a plurality of data terminals 114 via access terminal 112, wireless access 120 to a plurality of mobile devices 124 and vehicle 126 via base station or access point 122, voice access 130 to a plurality of telephony devices 134, via switching device 132 and/or media access 140 to a plurality of audio/video display devices 144 via media terminal 142. In addition, communication network 125 is coupled to one or more content sources 175 of audio, video, graphics, text and/or other media. While broadband access 110, wireless access 120, voice access 130 and media access 140 are shown separately, one or more of these forms of access can be combined to provide multiple access services to a single client device (e.g., mobile devices 124 can receive media content via media terminal 142, data terminal 114 can be provided voice access via switching device 132, and so on).

[0019] The communications network 125 includes a plurality of network elements (NE) 150, 152, 154, 156, etc. for facilitating the broadband access 110, wireless access 120, voice access 130, media access 140 and/or the distribution of content from content sources 175. The communications network 125 can include a circuit switched or packet switched network, a voice over Internet protocol (VOIP) network, Internet protocol (IP) network, a cable network, a passive or active optical network, a 4G, 5G, or higher generation wireless access network, WIMAX network, UltraWideband network, personal area network or other wireless access network, a broadcast satellite network and/or other communications network.

[0020] In various embodiments, the access terminal 112 can include a digital subscriber line access multiplexer (DSLAM), cable modem termination system (CMTS), optical line terminal (OLT) and/or other access terminal. The data terminals 114 can include personal computers, laptop computers, netbook computers, tablets or other computing devices along with digital subscriber line (DSL) modems, data over coax service interface specification (DOCSIS) modems or other cable modems, a wireless modem such as a 4G, 5G, or higher generation modem, an optical modem and/or other access devices.

[0021] In various embodiments, the base station or access point 122 can include a 4G, 5G, or higher generation base station, an access point that operates via an 802.11 standard such as 802.11n, 802.11ac or other wireless access terminal. The mobile devices 124 can include mobile phones, e-readers, tablets, phablets, wireless modems, and/or other mobile computing devices.

[0022] In various embodiments, the switching device 132 can include a private branch exchange or central office switch, a media services gateway, VoIP gateway or other gateway device and/or other switching device. The tele-

phony devices 134 can include traditional telephones (with or without a terminal adapter), VoIP telephones and/or other telephony devices.

[0023] In various embodiments, the media terminal 142 can include a cable head-end or other TV head-end, a satellite receiver, gateway or other media terminal 142. The display devices 144 can include televisions with or without a set top box, personal computers and/or other display devices.

[0024] In various embodiments, the content sources 175 include broadcast television and radio sources, video on demand platforms and streaming video and audio services platforms, one or more content data networks, data servers, web servers and other content servers, and/or other sources of media.

[0025] In various embodiments, the communications network 125 can include wired, optical and/or wireless links and the network elements 150, 152, 154, 156, etc. can include service switching points, signal transfer points, service control points, network gateways, media distribution hubs, servers, firewalls, routers, edge devices, switches and other network nodes for routing and controlling communications traffic over wired, optical and wireless links as part of the Internet and other public networks as well as one or more private networks, for managing subscriber access, for billing and network management and for supporting other network functions.

[0026] FIG. 2A is a block diagram illustrating an example, non-limiting embodiment of a system 200a in accordance with various aspects described herein. In some embodiments, the system 200a may function within, or may be operatively overlaid upon, the communication network 100 of FIG. 1. The system 200a may include one or more VNFs, such as for example a first VNF 202a-1, a second VNF 202a-2, and a third VNF 202a-3. In some embodiments, two or more of the VNFs 202a-1, 202a-2, and 202a-3 may correspond to instances or copies of a same/common VNF. As one skilled in the art will appreciate based on a review of this disclosure, utilizing a same/common VNF across a multitude of platforms may reduce the cost/expense associated with, e.g., developing, testing, validating, refining, troubleshooting, maintaining, etc., the VNF.

[0027] As used in this context, and as it relates to at least two VNFs, the two VNFs may be referred to as being the same/common while not necessarily being identical to one another. For example, subtle or minor changes may exist between the two VNFs. Two (or more) VNFs may be referred to herein as being a same/common VNF if they share at least some threshold amount (e.g., 80%, 85%, 90%, 95%, etc.) of characteristics in common with one another.

[0028] As shown in FIG. 2A, each of the VNFs 202a-1 through 202a-3 may be associated with a respective platform. For example, the first VNF 202a-1 may be associated with a first platform 206a-1, the second VNF 206a-2 may be associated with a second platform 206a-2, and the third VNF 202a-3 may be associated with a third platform 206a-3. Using conventional technologies/techniques, a given VNF is bound directly to a given platform.

[0029] The platforms 206a-1 through 206a-3 may be associated with respective, different cloud computing architectures and/or may be operated/managed by different providers. In this regard, and as alluded to above, each of the platforms 206a-1 through 206a-3 may utilize their own suite/set of configuration parameters, protocols, naming

conventions, etc. This implies that to implement a given VNF across two or more of the platforms may require the VNF to be substantially adapted/modified as between the two or more of the platforms using conventional techniques/technologies. However, and for at least the reasons set forth above, frequently it may be desirable to reduce (e.g., minimize) an amount of changes/modifications that are made to a VNF from one instance of the VNF to another instance of the VNF. In this respect, FIG. 2A illustrates an interface layer represented as a glider or glider layer (GL) being disposed between VNFs and platforms. For example, a first GL **210a-1** is shown as being disposed between the first VNF **202a-1** and the first platform **206a-1**, a second GL **210a-2** is shown as being disposed between the second VNF **202a-2** and the second platform **206a-2**, and a third GL **210a-3** is shown as being disposed between the third VNF **202a-3** and the third platform **206a-3**. A given GL (e.g., the first GL **210a-1**) may serve as an interface between its respective VNF (e.g., the first VNF **202a-1**) and the associated platform (e.g., the first platform **206a-1**). A GL may incorporate/include hardware, firmware, software, or any combination thereof, to facilitate the functionality of the GL as described in further detail below.

[0030] Two or more of the GLs shown in FIG. 2A may correspond to instances or copies of a same/common GL. A given GL may obtain data/information that may be needed to interface a VNF to a platform from a data repository **214a**, by way of a server **218a**. In some embodiments, the data repository **214a** and the server **218a** may be contained within a common housing or casing. While a server **218a** is shown in FIG. 2A, in some embodiments other types of communication devices (such as for example switches, routers, gateways, modems, base stations, user equipment, etc.) may be used. Based on the information/data that the given GL obtains from the server **218a**, the GL may map parameters/settings associated with the VNF to the (operating) parameters/settings associated with the respective platform. In this manner, the GL may assume responsibilities/obligations associated with interfacing the VNF to the platform, thereby potentially relieving the VNF and the platform (or respective, associated operators/providers) of such responsibilities/obligations.

[0031] The data repository **214a** may include data/information associated with the VNFs **202a-1** through **202a-3** and/or the platforms **206a-1** through **206a-3** to facilitate interfacing a given VNF to a given platform. The data repository **214a** may obtain or manage such data/information in respect of one or more local libraries, caches, memories, databases, technical manuals/specifications, or the like. In some embodiments, the data repository **214a** may obtain the data/information contained therein from one or more technical forums/groups, social media platforms, networks (e.g., the Internet), etc. In this regard, and to the extent that data/information that may be needed by the data repository **214a** is not locally available at the data repository **214a**, the data repository **214a** (or, analogously, the server **218a**) may process a setting/parameter associated with a VNF or platform via a natural language processing (NLP) technique to transform the setting/parameter to a (text) search string/query that may be utilized in connection with one or more search engines. The search string/query may, in turn and via a search engine, yield/generate search results that may be utilized to obtain the data/information that is needed. The data/information associated with the search

results may be stored at the data repository **214a** to facilitate future uses of the data/information.

[0032] Data/Information utilized by a GL (e.g., the first GL **210a-1**) may pertain to various types of parameters or settings, including but not limited to: transmission power levels, receiver sensitivity levels, data transfer rates, modulation schemes, demodulation schemes, security schemes (e.g., encryption/decryption schemes), authentication schemes, communication protocols, beam-forming/beam-steering (e.g., directionality of antennas), billing/invoicing functions, naming structures or functionalities, etc., or any combination thereof. In some embodiments, a GL may communicate with the server **218a** to share experiences/settings associated with a VNF or platform to build-up/refine the base of knowledge that is available at the server **218a** and/or the data repository **214a**.

[0033] As described above, in some embodiments it may be desirable to implement a given VNF across multiple platforms. For example, doing so may help to increase the reliability or availability of that VNF in the event of, e.g., platform down-time (e.g., scheduled or unscheduled service or maintenance in respect of a platform), excessive load (e.g., load in an amount greater than threshold) at a platform, changes in terms/conditions (e.g., an operator of a platform increasing the rate charged to an owner/operator of the VNF for hosting the VNF on the platform), etc.

[0034] To the extent that a given VNF is available at more than one platform (e.g., to the extent that instances of a given VNF are replicated at multiple platforms), the GL at a first of the platforms may serve as a primary GL and the GL(s) at the other platform(s) may function as (a) secondary GL(s). The primary GL may elect to relinquish/grant authority or responsibility for executing various functions or tasks to one or more of the secondary GLs. In some embodiments, authority for delegating functions or tasks to various GLs may be wholly or partially assigned to a controller; the controller may be implemented at the server **218a** in some embodiments.

[0035] Various considerations may be taken into account when deciding where to allocate tasks or functionality as between multiple GLs (or, analogously, as between multiple VNFs). For example, if on a given day the first platform **206a-1** charges a first rate for using the first VNF **202a-1** hosted thereon, the second platform **206a-2** charges a second rate for using the second VNF **202a-2** hosted thereon, and the third platform **206a-3** charges a third rate for using the third VNF **202a-3** hosted thereon, and assuming that the first rate is less than the second rate and the second rate is less than the third rate, and assuming that the VNFs are all responsible for facilitating communication services (e.g., communication sessions), all other conditions being assumed equal an operator or manager of the VNFs may elect to delegate responsibility for handling communication services on the given day to the first VNF **202a-1** in order to reduce the cost associated with the communication services. Cost is one consideration that may be taken into account; other considerations, such as for example quality of service (QoS) or quality of experience (QoE), load-distribution considerations (e.g., maintenance of load at a given VNF or platform being maintained at a value that is less than a threshold, potentially as a function of time), environmental impacts, etc., or any combination thereof, may be taken into account in some embodiments.

[0036] In some embodiments, an operator or manager of a VNF may elect to reduce the number of platforms that are used for hosting the VNF (for example, as part of cost-reduction operations). In this respect, and assuming that a VNF (e.g., VNF **202a-1**) on a first platform **206a-1** is intended to be eliminated in favor of a VNF (e.g., the VNF **202a-2**) on the second platform **206a-2**, the first GL **210a-1** may detect that the VNF **202a-1** is being ported to another platform (e.g., by virtue of a memory access, read, and copy operations having been detected by the first GL **210a-1**). The second GL **210a-2** associated with the second platform **206a-2** may detect the instantiation of the VNF at the second platform **206a-2** (e.g., by virtue of a memory access and write operations having been detected by the second GL **210a-2**). The second GL **210a-2** may communicate the instantiation of the VNF at the second platform **206a-2** to the first GL **210a-1** (such as for example if an address or other identifier of the first GL **210a-1** is available to the second GL **210a-2**) or another device (e.g., the server **218a**). Once the VNF is instantiated at the second platform **206a-2**, configuration or routing tables associated with the VNF may be updated/modified to map to the second platform **206a-2**, thereby potentially removing/deleting/destroying an instantiation of the VNF with respect to the first platform **206a-1** from such configuration/routing tables in the process. In some instances, the presence of the first platform **206a-1** in the configuration/routing tables may be retained for at least a threshold amount of time (e.g., one month) to ensure stability and successful operation of the VNF at the second platform **206a-2**; more generally, an expiration of a timer may be used to facilitate a deletion/removal of a platform in respect of an instantiation of a VNF. If the first GL **210a-1**, the second GL **210a-2**, and/or the server **218a** detects that the VNF is failing, in any respect, at the second platform **206a-2**, the VNF at the second platform **206a-2** may be at least temporarily disabled (potentially in favor of restoring the VNF at the first platform **206a-1**) until a repair of fix can be provided/generated to address the failure.

[0037] As the foregoing examples demonstrate, GLs may be communicatively coupled with one another (either directly, or via intermediate systems, devices, or components) to ensure continuous and seamless operations in respect of (instances of) one or more VNFs. Aspects of such communication coupling may provide for information sharing to facilitate redundancy, scalability, management, regulation, cost-reduction, and intelligent decision-making processes, while ensuring data integrity, privacy, and high levels of QoS/QoE.

[0038] Referring now to FIG. 2B, an illustrative embodiment of a method **200b** in accordance with various aspects described herein is shown. The method **200b** may be implemented (e.g., executed), in whole or in part, in conjunction with one or more systems, devices, and/or components, such as for example the systems, devices, and components set forth herein. Various operations of the method **200b** are described below in relation to the blocks shown in FIG. 2B.

[0039] In block **204b**, a VNF may be obtained. In this context, obtaining a VNF may refer to a creation/generation of at least a portion of the VNF in the first instance and/or accessing at least a portion of the VNF from a library, a database, a catalog, etc., in other instances.

[0040] In block **208b**, the VNF may be coupled to a GL. For example, as part of block **208b** the VNF may be embedded within the GL (or, analogously, the GL may be embedded within the VNF).

[0041] In block **212b**, the GL may be communicatively coupled with a platform and/or a server (or other communication device). As part of block **212b**, the platform may be selected. The selection may be based on one or more factors or considerations, such as the factors/considerations set forth above.

[0042] In block **216b**, the GL may obtain data/information to interface the VNF and the platform. Such data/information may be obtained via the server in the manner set forth above.

[0043] In block **220b**, the GL may translate protocols, messages, communications, etc., between the VNF and the platform utilizing the data/information obtained as part of block **216b**. In this respect, the VNF and the platform might not even necessarily realize that they are coupled to one another, which is to say that information/data pertaining to the interface between the VNF and the platform may be hidden from the VNF and/or the platform.

[0044] While for purposes of simplicity of explanation, the respective processes are shown and described as a series of blocks in FIG. 2B, it is to be understood and appreciated that the claimed subject matter is not limited by the order of the blocks, as some blocks may occur in different orders and/or concurrently with other blocks from what is depicted and described herein. Moreover, not all illustrated blocks may be required to implement the methods described herein.

[0045] In some embodiments, blocks or operations of the method **200b** may be replicated across multiple platforms. Stated slightly differently, aspects of the method **200b** may be instantiated in conjunction with one or more platforms to facilitate supporting a VNF across the one or more platforms.

[0046] Aspects of this disclosure may be utilized in conjunction with different types of services, such as for example communication services, content creation, generation, and/or distribution services, etc. In some embodiments, various criteria or parameters may be analyzed as part of decision-making processes or logic to facilitate such services.

[0047] Aspects of this disclosure may be utilized in connection with one or more network functions (NFs), such as for example one or more VNFs. In some embodiments, NFs may be instantiated in respect of one or more platforms, inclusive of one or more (cloud) computing platforms.

[0048] Aspects of this disclosure may be incorporated as part of one or more practical applications. For example, aspects of this disclosure may serve to reduce the cost and complexity associated with operating and managing a communication network. Based on principles of information sharing (or, analogously, information hiding), entities that require specific information may have access to that information while entities that do not require the specific information might not be burdened by/with the same. Aspects of this disclosure enable a VNF to be leveraged across multiple platforms without incurring the cost or burden associated with modifying the VNF from one platform to the next. In this regard, aspects of this disclosure represent substantial improvements in respect of the utilization and management of VNFs.

[0049] As described herein, various entities may be interfaced or coupled to one another. In some embodiments, a

coupling can include steps, actions, instructions and/or commands that initiate and/or maintain a communication connection between two or more entities.

[0050] Referring now to FIG. 3, a block diagram **300** is shown illustrating an example, non-limiting embodiment of a virtualized communication network in accordance with various aspects described herein. In particular a virtualized communication network is presented that can be used to implement some or all of the subsystems and functions of system **100**, the subsystems and functions of system **200a**, and method **200b** presented in FIGS. 1, 2A, and 2B. For example, virtualized communication network **300** can facilitate in whole or in part obtaining a VNF, coupling the VNF to an interface layer, selecting a computing platform from a plurality of computing platforms, resulting in a selected computing platform, coupling the interface layer to the selected computing platform, resulting in an instantiation of the VNF with respect to the selected computing platform, obtaining data to operatively interface the VNF and the selected computing platform, and translating, by the interface layer, a communication, a message, or a combination thereof, between the VNF and the selected computing platform using the data. Virtualized communication network **300** can facilitate in whole or in part coupling a VNF to a cloud computing platform, resulting in an instantiation of the VNF with respect to the cloud computing platform, obtaining, based on the coupling, data to operatively interface the VNF and the cloud computing platform, and translating a communication, a message, or a combination thereof, between the VNF and the cloud computing platform using the data. Virtualized communication network **300** can facilitate in whole or in part coupling a NF to a first computing platform that is operative in accordance with a first plurality of parameters, resulting in a first instantiation of the NF, coupling the NF to a second computing platform that is operative in accordance with a second plurality of parameters, resulting in a second instantiation of the NF, wherein the second plurality of parameters is at least partially different from the first plurality of parameters, obtaining first data to facilitate the first instantiation of the NF, and obtaining second data to facilitate the second instantiation of the NF.

[0051] In particular, a cloud networking architecture is shown that leverages cloud technologies and supports rapid innovation and scalability via a transport layer **350**, a virtualized network function cloud **325** and/or one or more cloud computing environments **375**. In various embodiments, this cloud networking architecture is an open architecture that leverages application programming interfaces (APIs); reduces complexity from services and operations; supports more nimble business models; and rapidly and seamlessly scales to meet evolving customer requirements including traffic growth, diversity of traffic types, and diversity of performance and reliability expectations.

[0052] In contrast to traditional network elements-which are typically integrated to perform a single function, the virtualized communication network employs virtual network elements (VNEs) **330**, **332**, **334**, etc. that perform some or all of the functions of network elements **150**, **152**, **154**, **156**, etc. For example, the network architecture can provide a substrate of networking capability, often called Network Function Virtualization Infrastructure (NFVI) or simply infrastructure that is capable of being directed with software and Software Defined Networking (SDN) proto-

cols to perform a broad variety of network functions and services. This infrastructure can include several types of substrates. The most typical type of substrate being servers that support Network Function Virtualization (NFV), followed by packet forwarding capabilities based on generic computing resources, with specialized network technologies brought to bear when general purpose processors or general purpose integrated circuit devices offered by merchants (referred to herein as merchant silicon) are not appropriate. In this case, communication services can be implemented as cloud-centric workloads.

[0053] As an example, a traditional network element **150** (shown in FIG. 1), such as an edge router can be implemented via a VNE **330** composed of NFV software modules, merchant silicon, and associated controllers. The software can be written so that increasing workload consumes incremental resources from a common resource pool, and moreover so that it's elastic: so the resources are only consumed when needed. In a similar fashion, other network elements such as other routers, switches, edge caches, and middle-boxes are instantiated from the common resource pool. Such sharing of infrastructure across a broad set of uses makes planning and growing infrastructure easier to manage.

[0054] In an embodiment, the transport layer **350** includes fiber, cable, wired and/or wireless transport elements, network elements and interfaces to provide broadband access **110**, wireless access **120**, voice access **130**, media access **140** and/or access to content sources **175** for distribution of content to any or all of the access technologies. In particular, in some cases a network element needs to be positioned at a specific place, and this allows for less sharing of common infrastructure. Other times, the network elements have specific physical layer adapters that cannot be abstracted or virtualized, and might require special DSP code and analog front-ends (AFEs) that do not lend themselves to implementation as VNEs **330**, **332** or **334**. These network elements can be included in transport layer **350**.

[0055] The virtualized network function cloud **325** interfaces with the transport layer **350** to provide the VNEs **330**, **332**, **334**, etc. to provide specific NFVs. In particular, the virtualized network function cloud **325** leverages cloud operations, applications, and architectures to support networking workloads. The virtualized network elements **330**, **332** and **334** can employ network function software that provides either a one-for-one mapping of traditional network element function or alternately some combination of network functions designed for cloud computing. For example, VNEs **330**, **332** and **334** can include route reflectors, domain name system (DNS) servers, and dynamic host configuration protocol (DHCP) servers, system architecture evolution (SAE) and/or mobility management entity (MME) gateways, broadband network gateways, IP edge routers for IP-VPN, Ethernet and other services, load balancers, distributors and other network elements. Because these elements don't typically need to forward large amounts of traffic, their workload can be distributed across a number of servers-each of which adds a portion of the capability, and overall which creates an elastic function with higher availability than its former monolithic version. These virtual network elements **330**, **332**, **334**, etc. can be instantiated and managed using an orchestration approach similar to those used in cloud compute services.

[0056] The cloud computing environments **375** can interface with the virtualized network function cloud **325** via

APIs that expose functional capabilities of the VNEs 330, 332, 334, etc. to provide the flexible and expanded capabilities to the virtualized network function cloud 325. In particular, network workloads may have applications distributed across the virtualized network function cloud 325 and cloud computing environment 375 and in the commercial cloud, or might simply orchestrate workloads supported entirely in NFV infrastructure from these third party locations.

[0057] Turning now to FIG. 4, there is illustrated a block diagram of a computing environment in accordance with various aspects described herein. In order to provide additional context for various embodiments of the embodiments described herein, FIG. 4 and the following discussion are intended to provide a brief, general description of a suitable computing environment 400 in which the various embodiments of the subject disclosure can be implemented. In particular, computing environment 400 can be used in the implementation of network elements 150, 152, 154, 156, access terminal 112, base station or access point 122, switching device 132, media terminal 142, and/or VNEs 330, 332, 334, etc. Each of these devices can be implemented via computer-executable instructions that can run on one or more computers, and/or in combination with other program modules and/or as a combination of hardware and software. For example, computing environment 400 can facilitate in whole or in part obtaining a VNF, coupling the VNF to an interface layer, selecting a computing platform from a plurality of computing platforms, resulting in a selected computing platform, coupling the interface layer to the selected computing platform, resulting in an instantiation of the VNF with respect to the selected computing platform, obtaining data to operatively interface the VNF and the selected computing platform, and translating, by the interface layer, a communication, a message, or a combination thereof, between the VNF and the selected computing platform using the data. Computing environment 400 can facilitate in whole or in part coupling a VNF to a cloud computing platform, resulting in an instantiation of the VNF with respect to the cloud computing platform, obtaining, based on the coupling, data to operatively interface the VNF and the cloud computing platform, and translating a communication, a message, or a combination thereof, between the VNF and the cloud computing platform using the data. Computing environment 400 can facilitate in whole or in part coupling a NF to a first computing platform that is operative in accordance with a first plurality of parameters, resulting in a first instantiation of the NF, coupling the NF to a second computing platform that is operative in accordance with a second plurality of parameters, resulting in a second instantiation of the NF, wherein the second plurality of parameters is at least partially different from the first plurality of parameters, obtaining first data to facilitate the first instantiation of the NF, and obtaining second data to facilitate the second instantiation of the NF.

[0058] Generally, program modules comprise routines, programs, components, data structures, etc., that perform particular tasks or implement particular abstract data types. Moreover, those skilled in the art will appreciate that the methods can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, minicomputers, mainframe computers, as well as personal computers, hand-held computing devices, microprocessor-based or programmable consumer

electronics, and the like, each of which can be operatively coupled to one or more associated devices.

[0059] As used herein, a processing circuit includes one or more processors as well as other application specific circuits such as an application specific integrated circuit, digital logic circuit, state machine, programmable gate array or other circuit that processes input signals or data and that produces output signals or data in response thereto. It should be noted that while any functions and features described herein in association with the operation of a processor could likewise be performed by a processing circuit.

[0060] The illustrated embodiments of the embodiments herein can be also practiced in distributed computing environments where certain tasks are performed by remote processing devices that are linked through a communications network. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0061] Computing devices typically comprise a variety of media, which can comprise computer-readable storage media and/or communications media, which two terms are used herein differently from one another as follows. Computer-readable storage media can be any available storage media that can be accessed by the computer and comprises both volatile and nonvolatile media, removable and non-removable media. By way of example, and not limitation, computer-readable storage media can be implemented in connection with any method or technology for storage of information such as computer-readable instructions, program modules, structured data or unstructured data.

[0062] Computer-readable storage media can comprise, but are not limited to, random access memory (RAM), read only memory (ROM), electrically erasable programmable read only memory (EEPROM), flash memory or other memory technology, compact disk read only memory (CD-ROM), digital versatile disk (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices or other tangible and/or non-transitory media which can be used to store desired information. In this regard, the terms “tangible” or “non-transitory” herein as applied to storage, memory or computer-readable media, are to be understood to exclude only propagating transitory signals per se as modifiers and do not relinquish rights to all standard storage, memory or computer-readable media that are not only propagating transitory signals per se.

[0063] Computer-readable storage media can be accessed by one or more local or remote computing devices, e.g., via access requests, queries or other data retrieval protocols, for a variety of operations with respect to the information stored by the medium.

[0064] Communications media typically embody computer-readable instructions, data structures, program modules or other structured or unstructured data in a data signal such as a modulated data signal, e.g., a carrier wave or other transport mechanism, and comprises any information delivery or transport media. The term “modulated data signal” or signals refers to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in one or more signals. By way of example, and not limitation, communication media comprise wired media, such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media.

[0065] With reference again to FIG. 4, the example environment can comprise a computer 402, the computer 402 comprising a processing unit 404, a system memory 406 and a system bus 408. The system bus 408 couples system components including, but not limited to, the system memory 406 to the processing unit 404. The processing unit 404 can be any of various commercially available processors. Dual microprocessors and other multiprocessor architectures can also be employed as the processing unit 404.

[0066] The system bus 408 can be any of several types of bus structure that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures. The system memory 406 comprises ROM 410 and RAM 412. A basic input/output system (BIOS) can be stored in a non-volatile memory such as ROM, erasable programmable read only memory (EPROM), EEPROM, which BIOS contains the basic routines that help to transfer information between elements within the computer 402, such as during startup. The RAM 412 can also comprise a high-speed RAM such as static RAM for caching data.

[0067] The computer 402 further comprises an internal hard disk drive (HDD) 414 (e.g., EIDE, SATA), which internal HDD 414 can also be configured for external use in a suitable chassis (not shown), a magnetic floppy disk drive (FDD) 416, (e.g., to read from or write to a removable diskette 418) and an optical disk drive 420, (e.g., reading a CD-ROM disk 422 or, to read from or write to other high capacity optical media such as the DVD). The HDD 414, magnetic FDD 416 and optical disk drive 420 can be connected to the system bus 408 by a hard disk drive interface 424, a magnetic disk drive interface 426 and an optical drive interface 428, respectively. The hard disk drive interface 424 for external drive implementations comprises at least one or both of Universal Serial Bus (USB) and Institute of Electrical and Electronics Engineers (IEEE) 1394 interface technologies. Other external drive connection technologies are within contemplation of the embodiments described herein.

[0068] The drives and their associated computer-readable storage media provide nonvolatile storage of data, data structures, computer-executable instructions, and so forth. For the computer 402, the drives and storage media accommodate the storage of any data in a suitable digital format. Although the description of computer-readable storage media above refers to a hard disk drive (HDD), a removable magnetic diskette, and a removable optical media such as a CD or DVD, it should be appreciated by those skilled in the art that other types of storage media which are readable by a computer, such as zip drives, magnetic cassettes, flash memory cards, cartridges, and the like, can also be used in the example operating environment, and further, that any such storage media can contain computer-executable instructions for performing the methods described herein.

[0069] A number of program modules can be stored in the drives and RAM 412, comprising an operating system 430, one or more application programs 432, other program modules 434 and program data 436. All or portions of the operating system, applications, modules, and/or data can also be cached in the RAM 412. The systems and methods described herein can be implemented utilizing various commercially available operating systems or combinations of operating systems.

[0070] A user can enter commands and information into the computer 402 through one or more wired/wireless input devices, e.g., a keyboard 438 and a pointing device, such as a mouse 440. Other input devices (not shown) can comprise a microphone, an infrared (IR) remote control, a joystick, a game pad, a stylus pen, touch screen or the like. These and other input devices are often connected to the processing unit 404 through an input device interface 442 that can be coupled to the system bus 408, but can be connected by other interfaces, such as a parallel port, an IEEE 1394 serial port, a game port, a universal serial bus (USB) port, an IR interface, etc.

[0071] A monitor 444 or other type of display device can be also connected to the system bus 408 via an interface, such as a video adapter 446. It will also be appreciated that in alternative embodiments, a monitor 444 can also be any display device (e.g., another computer having a display, a smart phone, a tablet computer, etc.) for receiving display information associated with computer 402 via any communication means, including via the Internet and cloud-based networks. In addition to the monitor 444, a computer typically comprises other peripheral output devices (not shown), such as speakers, printers, etc.

[0072] The computer 402 can operate in a networked environment using logical connections via wired and/or wireless communications to one or more remote computers, such as a remote computer(s) 448. The remote computer(s) 448 can be a workstation, a server computer, a router, a personal computer, portable computer, microprocessor-based entertainment appliance, a peer device or other common network node, and typically comprises many or all of the elements described relative to the computer 402, although, for purposes of brevity, only a remote memory/storage device 450 is illustrated. The logical connections depicted comprise wired/wireless connectivity to a local area network (LAN) 452 and/or larger networks, e.g., a wide area network (WAN) 454. Such LAN and WAN networking environments are commonplace in offices and companies, and facilitate enterprise-wide computer networks, such as intranets, all of which can connect to a global communications network, e.g., the Internet.

[0073] When used in a LAN networking environment, the computer 402 can be connected to the LAN 452 through a wired and/or wireless communication network interface or adapter 456. The adapter 456 can facilitate wired or wireless communication to the LAN 452, which can also comprise a wireless AP disposed thereon for communicating with the adapter 456.

[0074] When used in a WAN networking environment, the computer 402 can comprise a modem 458 or can be connected to a communications server on the WAN 454 or has other means for establishing communications over the WAN 454, such as by way of the Internet. The modem 458, which can be internal or external and a wired or wireless device, can be connected to the system bus 408 via the input device interface 442. In a networked environment, program modules depicted relative to the computer 402 or portions thereof, can be stored in the remote memory/storage device 450. It will be appreciated that the network connections shown are example and other means of establishing a communications link between the computers can be used.

[0075] The computer 402 can be operable to communicate with any wireless devices or entities operatively disposed in wireless communication, e.g., a printer, scanner, desktop

and/or portable computer, portable data assistant, communications satellite, any piece of equipment or location associated with a wirelessly detectable tag (e.g., a kiosk, news stand, restroom), and telephone. This can comprise Wireless Fidelity (Wi-Fi) and BLUETOOTH® wireless technologies. Thus, the communication can be a predefined structure as with a conventional network or simply an ad hoc communication between at least two devices.

[0076] Wi-Fi can allow connection to the Internet from a couch at home, a bed in a hotel room or a conference room at work, without wires. Wi-Fi is a wireless technology similar to that used in a cell phone that enables such devices, e.g., computers, to send and receive data indoors and out; anywhere within the range of a base station. Wi-Fi networks use radio technologies called IEEE 802.11 (a, b, g, n, ac, ag, etc.) to provide secure, reliable, fast wireless connectivity. A Wi-Fi network can be used to connect computers to each other, to the Internet, and to wired networks (which can use IEEE 802.3 or Ethernet). Wi-Fi networks operate in the unlicensed 2.4 and 5 GHz radio bands for example or with products that contain both bands (dual band), so the networks can provide real-world performance similar to the basic 10BaseT wired Ethernet networks used in many offices.

[0077] Turning now to FIG. 5, an embodiment **500** of a mobile network platform **510** is shown that is an example of network elements **150**, **152**, **154**, **156**, and/or VNEs **330**, **332**, **334**, etc. For example, platform **510** can facilitate in whole or in part obtaining a VNF, coupling the VNF to an interface layer, selecting a computing platform from a plurality of computing platforms, resulting in a selected computing platform, coupling the interface layer to the selected computing platform, resulting in an instantiation of the VNF with respect to the selected computing platform, obtaining data to operatively interface the VNF and the selected computing platform, and translating, by the interface layer, a communication, a message, or a combination thereof, between the VNF and the selected computing platform using the data. Platform **510** can facilitate in whole or in part coupling a VNF to a cloud computing platform, resulting in an instantiation of the VNF with respect to the cloud computing platform, obtaining, based on the coupling, data to operatively interface the VNF and the cloud computing platform, and translating a communication, a message, or a combination thereof, between the VNF and the cloud computing platform using the data. Platform **510** can facilitate in whole or in part coupling a NF to a first computing platform that is operative in accordance with a first plurality of parameters, resulting in a first instantiation of the NF, coupling the NF to a second computing platform that is operative in accordance with a second plurality of parameters, resulting in a second instantiation of the NF, wherein the second plurality of parameters is at least partially different from the first plurality of parameters, obtaining first data to facilitate the first instantiation of the NF, and obtaining second data to facilitate the second instantiation of the NF.

[0078] In one or more embodiments, the mobile network platform **510** can generate and receive signals transmitted and received by base stations or access points such as base station or access point **122**. Generally, mobile network platform **510** can comprise components, e.g., nodes, gateways, interfaces, servers, or disparate platforms, that facilitate both packet-switched (PS) (e.g., internet protocol (IP),

frame relay, asynchronous transfer mode (ATM)) and circuit-switched (CS) traffic (e.g., voice and data), as well as control generation for networked wireless telecommunication. As a non-limiting example, mobile network platform **510** can be included in telecommunications carrier networks, and can be considered carrier-side components as discussed elsewhere herein. Mobile network platform **510** comprises CS gateway node(s) **512** which can interface CS traffic received from legacy networks like telephony network(s) **540** (e.g., public switched telephone network (PSTN), or public land mobile network (PLMN)) or a signaling system #7 (SS7) network **560**. CS gateway node(s) **512** can authorize and authenticate traffic (e.g., voice) arising from such networks. Additionally, CS gateway node(s) **512** can access mobility, or roaming, data generated through SS7 network **560**; for instance, mobility data stored in a visited location register (VLR), which can reside in memory **530**. Moreover, CS gateway node(s) **512** interfaces CS-based traffic and signaling and PS gateway node(s) **518**. As an example, in a 3GPP UMTS network, CS gateway node(s) **512** can be realized at least in part in gateway GPRS support node(s) (GGSN). It should be appreciated that functionality and specific operation of CS gateway node(s) **512**, PS gateway node(s) **518**, and serving node(s) **516**, is provided and dictated by radio technology(ies) utilized by mobile network platform **510** for telecommunication over a radio access network **520** with other devices, such as a radiotelephone **575**.

[0079] In addition to receiving and processing CS-switched traffic and signaling, PS gateway node(s) **518** can authorize and authenticate PS-based data sessions with served mobile devices. Data sessions can comprise traffic, or content(s), exchanged with networks external to the mobile network platform **510**, like wide area network(s) (WANs) **550**, enterprise network(s) **570**, and service network(s) **580**, which can be embodied in local area network(s) (LANs), can also be interfaced with mobile network platform **510** through PS gateway node(s) **518**. It is to be noted that WANs **550** and enterprise network(s) **570** can embody, at least in part, a service network(s) like IP multimedia subsystem (IMS). Based on radio technology layer(s) available in technology resource(s) or radio access network **520**, PS gateway node(s) **518** can generate packet data protocol contexts when a data session is established; other data structures that facilitate routing of packetized data also can be generated. To that end, in an aspect, PS gateway node(s) **518** can comprise a tunnel interface (e.g., tunnel termination gateway (TTG) in 3GPP UMTS network(s) (not shown)) which can facilitate packetized communication with disparate wireless network(s), such as Wi-Fi networks.

[0080] In embodiment **500**, mobile network platform **510** also comprises serving node(s) **516** that, based upon available radio technology layer(s) within technology resource(s) in the radio access network **520**, convey the various packetized flows of data streams received through PS gateway node(s) **518**. It is to be noted that for technology resource(s) that rely primarily on CS communication, server node(s) can deliver traffic without reliance on PS gateway node(s) **518**; for example, server node(s) can embody at least in part a mobile switching center. As an example, in a 3GPP UMTS network, serving node(s) **516** can be embodied in serving GPRS support node(s) (SGSN).

[0081] For radio technologies that exploit packetized communication, server(s) **514** in mobile network platform **510**

can execute numerous applications that can generate multiple disparate packetized data streams or flows, and manage (e.g., schedule, queue, format . . .) such flows. Such application(s) can comprise add-on features to standard services (for example, provisioning, billing, customer support . . .) provided by mobile network platform 510. Data streams (e.g., content(s) that are part of a voice call or data session) can be conveyed to PS gateway node(s) 518 for authorization/authentication and initiation of a data session, and to serving node(s) 516 for communication thereafter. In addition to application server, server(s) 514 can comprise utility server(s), a utility server can comprise a provisioning server, an operations and maintenance server, a security server that can implement at least in part a certificate authority and firewalls as well as other security mechanisms, and the like. In an aspect, security server(s) secure communication served through mobile network platform 510 to ensure network's operation and data integrity in addition to authorization and authentication procedures that CS gateway node(s) 512 and PS gateway node(s) 518 can enact. Moreover, provisioning server(s) can provision services from external network(s) like networks operated by a disparate service provider; for instance, WAN 550 or Global Positioning System (GPS) network(s) (not shown). Provisioning server(s) can also provision coverage through networks associated to mobile network platform 510 (e.g., deployed and operated by the same service provider), such as the distributed antennas networks shown in FIG. 1(s) that enhance wireless service coverage by providing more network coverage.

[0082] It is to be noted that server(s) 514 can comprise one or more processors configured to confer at least in part the functionality of mobile network platform 510. To that end, the one or more processor can execute code instructions stored in memory 530, for example. It is should be appreciated that server(s) 514 can comprise a content manager, which operates in substantially the same manner as described hereinbefore.

[0083] In example embodiment 500, memory 530 can store information related to operation of mobile network platform 510. Other operational information can comprise provisioning information of mobile devices served through mobile network platform 510, subscriber databases; application intelligence, pricing schemes, e.g., promotional rates, flat-rate programs, couponing campaigns; technical specification(s) consistent with telecommunication protocols for operation of disparate radio, or wireless, technology layers; and so forth. Memory 530 can also store information from at least one of telephony network(s) 540, WAN 550, SS7 network 560, or enterprise network(s) 570. In an aspect, memory 530 can be, for example, accessed as part of a data store component or as a remotely connected memory store.

[0084] In order to provide a context for the various aspects of the disclosed subject matter, FIG. 5, and the following discussion, are intended to provide a brief, general description of a suitable environment in which the various aspects of the disclosed subject matter can be implemented. While the subject matter has been described above in the general context of computer-executable instructions of a computer program that runs on a computer and/or computers, those skilled in the art will recognize that the disclosed subject matter also can be implemented in combination with other program modules. Generally, program modules comprise

routines, programs, components, data structures, etc. that perform particular tasks and/or implement particular abstract data types.

[0085] Turning now to FIG. 6, an illustrative embodiment of a communication device 600 is shown. The communication device 600 can serve as an illustrative embodiment of devices such as data terminals 114, mobile devices 124, vehicle 126, display devices 144 or other client devices for communication via either communications network 125. For example, computing device 600 can facilitate in whole or in part obtaining a VNF, coupling the VNF to an interface layer, selecting a computing platform from a plurality of computing platforms, resulting in a selected computing platform, coupling the interface layer to the selected computing platform, resulting in an instantiation of the VNF with respect to the selected computing platform, obtaining data to operatively interface the VNF and the selected computing platform, and translating, by the interface layer, a communication, a message, or a combination thereof, between the VNF and the selected computing platform using the data. Computing device 600 can facilitate in whole or in part coupling a VNF to a cloud computing platform, resulting in an instantiation of the VNF with respect to the cloud computing platform, obtaining, based on the coupling, data to operatively interface the VNF and the cloud computing platform, and translating a communication, a message, or a combination thereof, between the VNF and the cloud computing platform using the data. Computing device 600 can facilitate in whole or in part coupling a NF to a first computing platform that is operative in accordance with a first plurality of parameters, resulting in a first instantiation of the NF, coupling the NF to a second computing platform that is operative in accordance with a second plurality of parameters, resulting in a second instantiation of the NF, wherein the second plurality of parameters is at least partially different from the first plurality of parameters, obtaining first data to facilitate the first instantiation of the NF, and obtaining second data to facilitate the second instantiation of the NF.

[0086] The communication device 600 can comprise a wireline and/or wireless transceiver 602 (herein transceiver 602), a user interface (UI) 604, a power supply 614, a location receiver 616, a motion sensor 618, an orientation sensor 620, and a controller 606 for managing operations thereof. The transceiver 602 can support short-range or long-range wireless access technologies such as Bluetooth®, ZigBee®, WiFi, DECT, or cellular communication technologies, just to mention a few (Bluetooth® and ZigBee® are trademarks registered by the Bluetooth® Special Interest Group and the ZigBee® Alliance, respectively). Cellular technologies can include, for example, CDMA-1X, UMTS/HSDPA, GSM/GPRS, TDMA/EDGE, EV/DO, WiMAX, SDR, LTE, as well as other next generation wireless communication technologies as they arise. The transceiver 602 can also be adapted to support circuit-switched wireline access technologies (such as PSTN), packet-switched wireline access technologies (such as TCP/IP, VoIP, etc.), and combinations thereof.

[0087] The UI 604 can include a depressible or touch-sensitive keypad 608 with a navigation mechanism such as a roller ball, a joystick, a mouse, or a navigation disk for manipulating operations of the communication device 600. The keypad 608 can be an integral part of a housing assembly of the communication device 600 or an indepen-

dent device operably coupled thereto by a tethered wireline interface (such as a USB cable) or a wireless interface supporting for example Bluetooth®. The keypad **608** can represent a numeric keypad commonly used by phones, and/or a QWERTY keypad with alphanumeric keys. The UI **604** can further include a display **610** such as monochrome or color LCD (Liquid Crystal Display), OLED (Organic Light Emitting Diode) or other suitable display technology for conveying images to an end user of the communication device **600**. In an embodiment where the display **610** is touch-sensitive, a portion or all of the keypad **608** can be presented by way of the display **610** with navigation features.

[0088] The display **610** can use touch screen technology to also serve as a user interface for detecting user input. As a touch screen display, the communication device **600** can be adapted to present a user interface having graphical user interface (GUI) elements that can be selected by a user with a touch of a finger. The display **610** can be equipped with capacitive, resistive or other forms of sensing technology to detect how much surface area of a user's finger has been placed on a portion of the touch screen display. This sensing information can be used to control the manipulation of the GUI elements or other functions of the user interface. The display **610** can be an integral part of the housing assembly of the communication device **600** or an independent device communicatively coupled thereto by a tethered wireline interface (such as a cable) or a wireless interface.

[0089] The UI **604** can also include an audio system **612** that utilizes audio technology for conveying low volume audio (such as audio heard in proximity of a human ear) and high volume audio (such as speakerphone for hands free operation). The audio system **612** can further include a microphone for receiving audible signals of an end user. The audio system **612** can also be used for voice recognition applications. The UI **604** can further include an image sensor **613** such as a charged coupled device (CCD) camera for capturing still or moving images.

[0090] The power supply **614** can utilize common power management technologies such as replaceable and rechargeable batteries, supply regulation technologies, and/or charging system technologies for supplying energy to the components of the communication device **600** to facilitate long-range or short-range portable communications. Alternatively, or in combination, the charging system can utilize external power sources such as DC power supplied over a physical interface such as a USB port or other suitable tethering technologies.

[0091] The location receiver **616** can utilize location technology such as a global positioning system (GPS) receiver capable of assisted GPS for identifying a location of the communication device **600** based on signals generated by a constellation of GPS satellites, which can be used for facilitating location services such as navigation. The motion sensor **618** can utilize motion sensing technology such as an accelerometer, a gyroscope, or other suitable motion sensing technology to detect motion of the communication device **600** in three-dimensional space. The orientation sensor **620** can utilize orientation sensing technology such as a magnetometer to detect the orientation of the communication device **600** (north, south, west, and east, as well as combined orientations in degrees, minutes, or other suitable orientation metrics).

[0092] The communication device **600** can use the transceiver **602** to also determine a proximity to a cellular, WiFi, Bluetooth®, or other wireless access points by sensing techniques such as utilizing a received signal strength indicator (RSSI) and/or signal time of arrival (TOA) or time of flight (TOF) measurements. The controller **606** can utilize computing technologies such as a microprocessor, a digital signal processor (DSP), programmable gate arrays, application specific integrated circuits, and/or a video processor with associated storage memory such as Flash, ROM, RAM, SRAM, DRAM or other storage technologies for executing computer instructions, controlling, and processing data supplied by the aforementioned components of the communication device **600**.

[0093] Other components not shown in FIG. 6 can be used in one or more

[0094] embodiments of the subject disclosure. For instance, the communication device **600** can include a slot for adding or removing an identity module such as a Subscriber Identity Module (SIM) card or Universal Integrated Circuit Card (UICC). SIM or UICC cards can be used for identifying subscriber services, executing programs, storing subscriber data, and so on.

[0095] The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is for clarity only and doesn't otherwise indicate or imply any order in time. For instance, “a first determination,” “a second determination,” and “a third determination,” does not indicate or imply that the first determination is to be made before the second determination, or vice versa, etc.

[0096] In the subject specification, terms such as “store,” “storage,” “data store,” data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components described herein can be either volatile memory or nonvolatile memory, or can comprise both volatile and nonvolatile memory, by way of illustration, and not limitation, volatile memory, non-volatile memory, disk storage, and memory storage. Further, non-volatile memory can be included in read only memory (ROM), programmable ROM (PROM), electrically programmable ROM (EPROM), electrically erasable ROM (EEPROM), or flash memory. Volatile memory can comprise random access memory (RAM), which acts as external cache memory. By way of illustration and not limitation, RAM is available in many forms such as synchronous RAM (SRAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), double data rate SDRAM (DDR SDRAM), enhanced SDRAM (ESDRAM), Synchlink DRAM (SL-DRAM), and direct Rambus RAM (DRRAM). Additionally, the disclosed memory components of systems or methods herein are intended to comprise, without being limited to comprising, these and any other suitable types of memory.

[0097] Moreover, it will be noted that the disclosed subject matter can be practiced with other computer system configurations, comprising single-processor or multiprocessor computer systems, mini-computing devices, mainframe computers, as well as personal computers, hand-held computing devices (e.g., PDA, phone, smartphone, watch, tablet computers, netbook computers, etc.), microprocessor-based or programmable consumer or industrial electronics, and the like. The illustrated aspects can also be practiced in distrib-

uted computing environments where tasks are performed by remote processing devices that are linked through a communications network; however, some if not all aspects of the subject disclosure can be practiced on stand-alone computers. In a distributed computing environment, program modules can be located in both local and remote memory storage devices.

[0098] In one or more embodiments, information regarding use of services can be generated including services being accessed, media consumption history, user preferences, and so forth. This information can be obtained by various methods including user input, detecting types of communications (e.g., video content vs. audio content), analysis of content streams, sampling, and so forth. The generating, obtaining and/or monitoring of this information can be responsive to an authorization provided by the user. In one or more embodiments, an analysis of data can be subject to authorization from user(s) associated with the data, such as an opt-in, an opt-out, acknowledgement requirements, notifications, selective authorization based on types of data, and so forth.

[0099] Some of the embodiments described herein can also employ artificial intelligence (AI) to facilitate automating one or more features described herein. The embodiments (e.g., in connection with automatically identifying acquired cell sites that provide a maximum value/benefit after addition to an existing communication network) can employ various AI-based schemes for carrying out various embodiments thereof. Moreover, the classifier can be employed to determine a ranking or priority of each cell site of the acquired network. A classifier is a function that maps an input attribute vector, $X=(x_1, x_2, x_3, x_4, \dots, x_n)$, to a confidence that the input belongs to a class, that is, $f(x)=\text{confidence}(\text{class})$. Such classification can employ a probabilistic and/or statistical-based analysis (e.g., factoring into the analysis utilities and costs) to determine or infer an action that a user desires to be automatically performed. A support vector machine (SVM) is an example of a classifier that can be employed. The SVM operates by finding a hypersurface in the space of possible inputs, which the hypersurface attempts to split the triggering criteria from the non-triggering events. Intuitively, this makes the classification correct for testing data that is near, but not identical to training data. Other directed and undirected model classification approaches comprise, e.g., naïve Bayes, Bayesian networks, decision trees, neural networks, fuzzy logic models, and probabilistic classification models providing different patterns of independence can be employed. Classification as used herein also is inclusive of statistical regression that is utilized to develop models of priority.

[0100] As will be readily appreciated, one or more of the embodiments can employ classifiers that are explicitly trained (e.g., via a generic training data) as well as implicitly trained (e.g., via observing UE behavior, operator preferences, historical information, receiving extrinsic information). For example, SVMs can be configured via a learning or training phase within a classifier constructor and feature selection module. Thus, the classifier(s) can be used to automatically learn and perform a number of functions, including but not limited to determining according to pre-determined criteria which of the acquired cell sites will benefit a maximum number of subscribers and/or which of the acquired cell sites will add minimum value to the existing communication network coverage, etc.

[0101] As used in some contexts in this application, in some embodiments, the terms “component,” “system” and the like are intended to refer to, or comprise, a computer-related entity or an entity related to an operational apparatus with one or more specific functionalities, wherein the entity can be either hardware, a combination of hardware and software, software, or software in execution. As an example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an executable, a thread of execution, computer-executable instructions, a program, and/or a computer. By way of illustration and not limitation, both an application running on a server and the server can be a component. One or more components may reside within a process and/or thread of execution and a component may be localized on one computer and/or distributed between two or more computers. In addition, these components can execute from various computer readable media having various data structures stored thereon. The components may communicate via local and/or remote processes such as in accordance with a signal having one or more data packets (e.g., data from one component interacting with another component in a local system, distributed system, and/or across a network such as the Internet with other systems via the signal). As another example, a component can be an apparatus with specific functionality provided by mechanical parts operated by electric or electronic circuitry, which is operated by a software or firmware application executed by a processor, wherein the processor can be internal or external to the apparatus and executes at least a part of the software or firmware application. As yet another example, a component can be an apparatus that provides specific functionality through electronic components without mechanical parts, the electronic components can comprise a processor therein to execute software or firmware that confers at least in part the functionality of the electronic components. While various components have been illustrated as separate components, it will be appreciated that multiple components can be implemented as a single component, or a single component can be implemented as multiple components, without departing from example embodiments.

[0102] Further, the various embodiments can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware or any combination thereof to control a computer to implement the disclosed subject matter. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable device or computer-readable storage/communications media. For example, computer readable storage media can include, but are not limited to, magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips), optical disks (e.g., compact disk (CD), digital versatile disk (DVD)), smart cards, and flash memory devices (e.g., card, stick, key drive). Of course, those skilled in the art will recognize many modifications can be made to this configuration without departing from the scope or spirit of the various embodiments.

[0103] In addition, the words “example” and “exemplary” are used herein to mean serving as an instance or illustration. Any embodiment or design described herein as “example” or “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. Rather, use of the word example or exemplary is intended to

present concepts in a concrete fashion. As used in this application, the term “or” is intended to mean an inclusive “or” rather than an exclusive “or”. That is, unless specified otherwise or clear from context, “X employs A or B” is intended to mean any of the natural inclusive permutations. That is, if X employs A; X employs B; or X employs both A and B, then “X employs A or B” is satisfied under any of the foregoing instances. In addition, the articles “a” and “an” as used in this application and the appended claims should generally be construed to mean “one or more” unless specified otherwise or clear from context to be directed to a singular form.

[0104] Moreover, terms such as “user equipment,” “mobile station,” “mobile,” “subscriber station,” “access terminal,” “terminal,” “handset,” “mobile device” (and/or terms representing similar terminology) can refer to a wireless device utilized by a subscriber or user of a wireless communication service to receive or convey data, control, voice, video, sound, gaming or substantially any data-stream or signaling-stream. The foregoing terms are utilized interchangeably herein and with reference to the related drawings.

[0105] Furthermore, the terms “user,” “subscriber,” “customer,” “consumer” and the like are employed interchangeably throughout, unless context warrants particular distinctions among the terms. It should be appreciated that such terms can refer to human entities or automated components supported through artificial intelligence (e.g., a capacity to make inference based, at least, on complex mathematical formalisms), which can provide simulated vision, sound recognition and so forth.

[0106] As employed herein, the term “processor” can refer to substantially any computing processing unit or device comprising, but not limited to comprising, single-core processors; single-processors with software multithread execution capability; multi-core processors; multi-core processors with software multithread execution capability; multi-core processors with hardware multithread technology; parallel platforms; and parallel platforms with distributed shared memory. Additionally, a processor can refer to an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field programmable gate array (FPGA), a programmable logic controller (PLC), a complex programmable logic device (CPLD), a discrete gate or transistor logic, discrete hardware components or any combination thereof designed to perform the functions described herein. Processors can exploit nano-scale architectures such as, but not limited to, molecular and quantum-dot based transistors, switches and gates, in order to optimize space usage or enhance performance of user equipment. A processor can also be implemented as a combination of computing processing units.

[0107] As used herein, terms such as “data storage,” “data storage,” “database,” and substantially any other information storage component relevant to operation and functionality of a component, refer to “memory components,” or entities embodied in a “memory” or components comprising the memory. It will be appreciated that the memory components or computer-readable storage media, described herein can be either volatile memory or nonvolatile memory or can include both volatile and nonvolatile memory.

[0108] What has been described above includes mere examples of various embodiments. It is, of course, not possible to describe every conceivable combination of com-

ponents or methodologies for purposes of describing these examples, but one of ordinary skill in the art can recognize that many further combinations and permutations of the present embodiments are possible. Accordingly, the embodiments disclosed and/or claimed herein are intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the appended claims. Furthermore, to the extent that the term “includes” is used in either the detailed description or the claims, such term is intended to be inclusive in a manner similar to the term “comprising” as “comprising” is interpreted when employed as a transitional word in a claim.

[0109] In addition, a flow diagram may include a “start” and/or “continue” indication. The “start” and “continue” indications reflect that the steps presented can optionally be incorporated in or otherwise used in conjunction with other routines. In this context, “start” indicates the beginning of the first step presented and may be preceded by other activities not specifically shown. Further, the “continue” indication reflects that the steps presented may be performed multiple times and/or may be succeeded by other activities not specifically shown. Further, while a flow diagram indicates a particular ordering of steps, other orderings are likewise possible provided that the principles of causality are maintained.

[0110] As may also be used herein, the term(s) “operably coupled to”, “coupled to”, and/or “coupling” includes direct coupling between items and/or indirect coupling between items via one or more intervening items. Such items and intervening items include, but are not limited to, junctions, communication paths, components, circuit elements, circuits, functional blocks, and/or devices. As an example of indirect coupling, a signal conveyed from a first item to a second item may be modified by one or more intervening items by modifying the form, nature or format of information in a signal, while one or more elements of the information in the signal are nevertheless conveyed in a manner than can be recognized by the second item. In a further example of indirect coupling, an action in a first item can cause a reaction on the second item, as a result of actions and/or reactions in one or more intervening items.

[0111] Although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement which achieves the same or similar purpose may be substituted for the embodiments described or shown by the subject disclosure. The subject disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, can be used in the subject disclosure. For instance, one or more features from one or more embodiments can be combined with one or more features of one or more other embodiments. In one or more embodiments, features that are positively recited can also be negatively recited and excluded from the embodiment with or without replacement by another structural and/or functional feature. The steps or functions described with respect to the embodiments of the subject disclosure can be performed in any order. The steps or functions described with respect to the embodiments of the subject disclosure can be performed alone or in combination with other steps or functions of the subject disclosure, as well as from other embodiments or from other steps that have not been described in the subject disclosure. Further,

more than or less than all of the features described with respect to an embodiment can also be utilized.

What is claimed is:

1. A device, comprising:
 - a processing system including a processor; and
 - a memory that stores executable instructions that, when executed by the processing system, facilitate performance of operations, the operations comprising:
 - obtaining a virtual network function (VNF);
 - coupling the VNF to an interface layer;
 - coupling the interface layer to a computing platform, resulting in an instantiation of the VNF with respect to the computing platform;
 - obtaining data to operatively interface the VNF and the computing platform; and
 - translating, by the interface layer, a communication, a message, or a combination thereof, between the VNF and the computing platform using the data,
 wherein the obtaining of the data includes obtaining first data of the data from a technical forum, obtaining second data of the data from a social media platform, and obtaining third data of the data from a network.
2. The device of claim 1, wherein the network includes the Internet.
3. The device of claim 1, wherein the operations further comprise:
 - processing a setting associated with the VNF to transform the setting to a search query; and
 - processing the search query to generate search results, wherein the obtaining of the data is based on the search results.
4. The device of claim 1, wherein the obtaining of the VNF comprises generating a first portion of the VNF.
5. The device of claim 4, wherein the obtaining of the VNF comprises obtaining a second portion of the VNF from a library, a catalog, or a database of VNFs.
6. The device of claim 1, wherein the operations further comprise:
 - instantiating the VNF with respect to a second computing platform.
7. The device of claim 6, wherein the instantiating of the VNF with respect to the second computing platform comprises:
 - coupling the VNF to a second interface layer; and
 - coupling the second interface layer to the second computing platform.
8. The device of claim 7, wherein the instantiating of the VNF with respect to the second computing platform further comprises:
 - obtaining fourth data to operatively interface the VNF and the second computing platform.
9. The device of claim 8, wherein the fourth data is at least partially different from the data.
10. The device of claim 9, wherein the operations further comprise:
 - translating, by the second interface layer, a second communication, a second message, or a combination thereof, between the VNF and the second computing platform using the fourth data.
11. The device of claim 6, wherein the operations further comprise:
 - subsequent to the instantiating of the VNF with respect to the second computing platform, deleting the instantiation of the VNF with respect to the computing platform.

12. The device of claim 11, wherein the deleting of the instantiation of the VNF with respect to the computing platform occurs following an expiration of a timer relative to the instantiating of the VNF with respect to the second computing platform.

13. The device of claim 6, wherein the operations further comprise:

subsequent to the instantiating of the VNF with respect to the second computing platform, conveying first network traffic utilizing the instantiation of the VNF with respect to the computing platform and conveying second network traffic utilizing the instantiation of the VNF with respect to the second computing platform.

14. The device of claim 1, wherein the operations further comprise:

selecting the computing platform from a plurality of computing platforms based on a quality of service parameter, a quality of experience parameter, or a combination thereof, applicable to a user device in respect of a communication service, a content distribution service, or a combination thereof.

15. A non-transitory machine-readable medium, comprising executable instructions that, when executed by a processing system including a processor, facilitate performance of operations, the operations comprising:

- obtaining a virtual network function (VNF);
- coupling the VNF to a computing platform, resulting in an instantiation of the VNF with respect to the computing platform;
- obtaining data to operatively interface the VNF and the computing platform; and
- translating, by an interface layer, a communication, a message, or a combination thereof, between the VNF and the computing platform using the data,

wherein the obtaining of the data includes obtaining first data of the data from a technical forum, obtaining second data of the data from a social media platform, and obtaining third data of the data from a network.

16. The non-transitory machine-readable medium of claim 15, wherein the operations further comprise:

- determining that fourth data of the data is available in a local storage of the processing system; and
- responsive to the determining, obtaining the fourth data from the local storage.

17. The non-transitory machine-readable medium of claim 15, wherein the operations further comprise:

- determining that fourth data of the data is unavailable in a local storage of the processing system;
- responsive to the determining, translating one or more parameters of the VNF to a search query;
- submitting the search query via a search engine to obtain search results; and
- generating the fourth data from the search results.

18. The non-transitory machine-readable medium of claim 17, wherein the operations further comprise:

- storing the fourth data to the local storage based on the generating of the fourth data.

19. A method, comprising:

- obtaining, by a processing system including a processor, a network function (NF);
- coupling, by the processing system, the NF to an interface layer;

coupling, by the processing system, the interface layer to a computing platform, resulting in an instantiation of the NF with respect to the computing platform; obtaining, by the processing system, data to operatively interface the NF and the computing platform; and translating, by the processing system and via the interface layer, a communication and a message between the NF and the computing platform using the data, wherein the obtaining of the data includes obtaining first data of the data from a technical forum, obtaining second data of the data from a social media platform, and obtaining third data of the data from a network.

20. The method of claim **19**, further comprising: subsequent to the instantiation of the NF, determining, by the processing system, that a timer has expired; and removing, by the processing system and based on the determining that the timer has expired, the coupling of the interface layer to the computing platform.

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